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Enhancing Students' Access to Scholarship and Competition Opportunities: A Web-Based Platform for Search and Application

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Abstract—Access to scholarship and competition information for students in Indonesia is often hindered by decentralized data and limited accessibility. This research introduces a centralized web-based architecture engineered to aggregate diverse educational opportunities within a unified digital ecosystem. The proposed platform implements an intelligent recommendation engine that leverages Natural Language Processing (NLP) and BM25-based retrieval to identify relevant scholarship and competition opportunities according to students' profiles, interests, and educational backgrounds. Furthermore, the system incorporates an "Opportunity Strategy Suggestion" feature to assist students in portfolio development. Preliminary development indicates that the platform can streamline the search process and provide strategic guidance for secondary school students in their academic journey.

Keywords—BM25, scholarship recommendation, competition recommendation, information retrieval, natural language processing, web platform

I. INTRODUCTION

Access to scholarship and competition information remains a challenge for many students in Indonesia due to fragmented information sources and uneven dissemination mechanisms. Students often rely on social media, school announcements, or organizer websites to search for opportunities, making the process inefficient and inconsistent [1], [2]. In many scholarship programs, students are also required to demonstrate academic or non-academic achievements, which makes participation in competitions an important supporting factor [3], [4]. However, many students still experience difficulties in identifying relevant competitions and scholarship opportunities because information is scattered across multiple platforms and is not always verified or well organized [5], [6]. This issue becomes more noticeable in schools outside major cities, where information sharing still depends heavily on manual communication channels [7].

To address these challenges, this research proposes a web-based platform that integrates scholarship and competition information into a centralized system [8], [9]. The platform implements an AI-assisted recommendation mechanism to provide more personalized opportunities based on students' interests, talents, and educational backgrounds [10], [11]. In addition, the proposed system introduces an Opportunity Strategy Suggestion feature that helps students identify competitions and activities that may support their future scholarship preparation [12]. Unlike conventional platforms that mainly focus on information dissemination, the proposed system emphasizes personalized recommendations and strategic portfolio development for junior and senior high school students.

II. RELATED WORK

Previous studies have explored scholarship recommendation systems, educational decision-support systems, and AI-based recommendation approaches in academic environments. However, most existing platforms mainly focus on university students, while systems specifically designed for junior and senior high school students remain limited [14], [15]. In practice, many secondary school students still depend on fragmented information sources such as social media, school announcements, and organizer websites, which often leads to inefficient information discovery and missed opportunities [16], [17]. Research in educational recommendation systems also shows that decision-support and profile-based recommendation approaches can improve recommendation relevance by considering users' interests, academic backgrounds, and educational needs [18], [19].

Recent developments in recommendation systems indicate that retrieval-based and hybrid recommendation approaches can improve personalization and

recommendation relevance in educational platforms [20], [21]. Nevertheless, most current systems still focus primarily on information dissemination and provide limited support for long-term portfolio planning and personalized opportunity guidance. In addition, many existing platforms do not sufficiently consider the specific characteristics of secondary school students, such as educational level, talents, interests, and achievement profiles. These limitations highlight the need for a centralized and AI-assisted platform capable of providing more relevant scholarship and competition recommendations for secondary school students.

III. METHOD

This study implements a digital ecosystem that simplifies scholarship and competition discovery for students. To automate that discovery process, we combined software engineering practices with natural language processing (NLP).

A. Systematic Research Framework

To realize our concept as a digital solution, we defined a six-step process. First, we established the primary goal: linking scholarships and competitions within a single system. Second, we analyzed the problems students encounter when manually searching for such information. Third, we designed the system architecture. Fourth, we implemented that design—building the core infrastructure and the AI modules. Fifth, we tested the integrated system. Sixth, we evaluated the outcomes to confirm that the centralized hub delivers relevant, easy-to-use information that effectively supports students.

B. Data Acquisition and Problem Validation

To ensure practical utility, we collected data from multiple sources. First, we examined existing educational aggregators to identify their shortcomings. Second, we observed how students manually search for information; this revealed significant time loss due to information being scattered across the internet. Third, we conducted direct interviews with students. A key finding emerged: many students participate in competitions without realizing that those achievements could qualify them for scholarships.

C. Technical Architecture and System Tiers

The platform implements a decoupled three-tier architecture. This design keeps the user interface, business logic, and data storage separate. On the front end, the presentation layer uses CSS, and JavaScript together with the VILT framework; as a result, the interface remains responsive across different devices. For the system core, a Go-based backend handles high-performance processing. Laravel works alongside Go to provide robust security and database orchestration. Meanwhile, Vue.js for front end and Inertia for bridges between the front end and the back end. All critical data including scholarship listings and competition records reside in a Postgree Repository and SQLite.

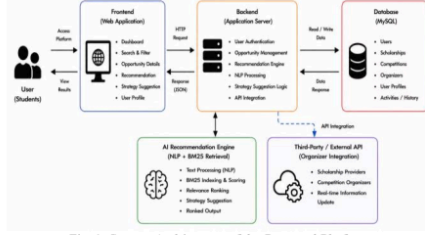


Fig. 1. System Architecture of the Proposed Platform.

Figure 1 illustrates the proposed system architecture, showing the interaction between the user interface, backend services, database, AI recommendation engine, and external organizer APIs. This architecture supports centralized information access and personalized opportunity recommendations.

D. Intelligent Recommendation Engine

The recommendation engine serves as the core component of the proposed platform, responsible for matching student profiles with the most relevant scholarship and competition opportunities. Instead of relying solely on conventional keyword-frequency approaches, the system applies the BM25 retrieval model to rank opportunities according to their relevance to user profiles.

BM25 is a probabilistic ranking algorithm commonly used in information retrieval systems because it considers both term importance and document length normalization. This allows the recommendation process to produce more accurate and context-aware results compared to traditional TF-IDF weighting approaches.

In the proposed system, student interests, educational backgrounds, talents, and preferred activity categories are processed using Natural Language Processing (NLP). The processed profile information is then compared against scholarship and competition descriptions stored in the database. BM25 scoring is used to calculate the relevance between user queries and opportunity documents.

The BM25 relevance score is computed using the following equation:

$$BM25(q, d) = \sum_{i=1}^n \frac{f(q_i, d) \cdot (k_1 + 1) \cdot f(q_i, d) + k_1(1 - b + b \cdot \frac{d}{avgdl})}{f(q_i, d) \cdot (k_1 + 1) \cdot f(q_i, d) + k_1(1 - b + b \cdot \frac{d}{avgdl})}$$

where:

- q represents the processed user profile query,
- d represents the opportunity document,
- f_i, d indicates the frequency of term i in document d ,
- $|d|$ denotes document length,
- $avgdl$ refers to the average document length in the collection,
- k_1 and b are tuning parameters used for frequency saturation and length normalization.

By applying BM25 scoring, the system ranks available opportunities and returns the most relevant recommendations according to students' characteristics and educational goals.

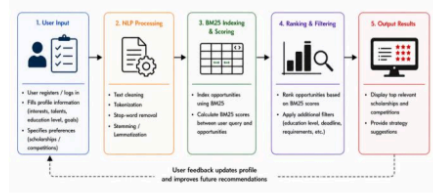


Fig. 2. Recommendation Workflow of the Proposed System.

E. Experimental Protocol and Evaluation

To see if our method works we tested the system with two groups: high school seniors and first-year university students. We looked at three things:

1. How accurately the AI matched students, with eligible scholarships.
2. How stable the system was when many users were using it at the time.
3. How much time students saved by using our platform compared to methods.

IV. RESULTS

The results of the system implementation were evaluated based on three main research questions: First, determine the accuracy of the AI recommendation system in matching relevant competitions and scholarships to the student profile. Second, assess the stability of the system when used by many users at the same time. Third, compare the time efficiency gained by students when using the system compared to manual information search methods.

A. Recommendation System Evaluation

The proposed recommendation system adopts a content-based retrieval approach to match student profiles with relevant scholarship and competition opportunities. BM25 is used as the primary retrieval and ranking method, while TF-IDF with cosine similarity and Jaccard similarity serves as the baseline comparison. This setup is used to measure how well the system identifies opportunities that align with students' interests, educational backgrounds, talents, and achievement profiles.

For evaluation, a dataset of more than 100 synthetic scholarship and competition records was prepared to reflect opportunities commonly available to Indonesian students. The dataset was split into 85% training data and 15% testing data. Performance was measured using precision, recall, F1-score, and Mean Reciprocal Rank (MRR) to assess relevance, ranking quality, and consistency across the compared methods.

TABLE I
TEST PROFILES USED FOR EVALUATION

Student Profile	Level	Scholarship Path	Interest
A high-achieving student with active participation in academic Olympiads.	High School	STEM	Science
A Creative student with talent on arts in need for general scholarships to support her financial.	High School	Art	General
A science and athletics student who is interested in scholarship through sports achievements.	Junior High School	Sports	Science

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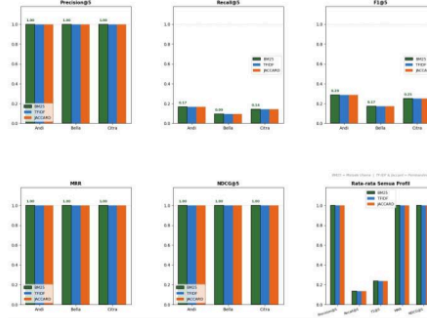


Fig. 3. Performance Comparison Between BM25, TF-IDF, and Jaccard Similarity.

As shown in Figure 3, the values produced by BM25, TF-IDF, and JACCARD are similar in all metrics used for comparison. These metrics include Precision@5 of 1.00, Recall@5 of 0.135, F1-Score@5 of 0.236, MRR of 1.00, and NDCG@5 of 1.00. This demonstrates that the three methods possess an equivalent capacity to generate scholarship and olympiad recommendations that are pertinent to the student profile on the dataset utilized.

The similarity of these results may be due to several factors. First, the dataset's limited size means the keyword distribution among items is not complex enough to produce significant differences between methods. Second, the test profile has clear and specific attributes, enabling the identification of relevant items by all methods. Third, the ground truth is manually determined based on simple attribute suitability rules, so different methods tend to produce similar rankings. Therefore, even though there was no numerical difference in performance, these results prove that BM25 can produce consistent and accurate recommendations.

Although BM25, TF-IDF, and Jaccard performed similarly in this test, BM25 was selected as the primary method in this study. This selection is based on BM25's theoretical advantages in normalizing word frequency and document

length, which make it more robust when faced with larger datasets and more varied descriptions of opportunities. TF-IDF and Jaccard are used as baseline methods to validate the consistency of the recommendation results.

TABLE II
THE RESULT OF PERSONALIZED RECOMMENDATION OUTPUTS

Student Profile	Dominant Olympiad Recommendation	Dominant Scholarship Recommendation
Active Academic Olympiads student	Science olympiads and STEM academic competitions	STEM-focused scholarships
Creative student	Art and Creative Competitions	General Scholastships
Sports student that has interest in science	Sports Competitions (O2SN)	Sports Scholarships

Table II summarizes the recommendation results generated by the system for each test profile. For Active academic olympiads student and Creative Student, the recommendation system has accomodate based on their needs. But for the Sport Student, the recommendation result still dominated by sports-related competitions and scholarships, despite the student's interest in science. This means that the system were dominantly ocured based of the scholarship path attributed while in the matching process. As a result, the system prioritizes opportunities aligned with the target scholarship pathway rather than the student's academic interest. And this might reduce recommendation diversity in profiles with conflicting characteristics.

In conclusion, this study still has several limitations. First, the dataset and user profiles used for the evaluation were synthetic and may not accurately reflect real-world user preferences. Second, the limited number of test profiles resulted in a relatively narrow range of evaluation scenarios. Third, the current weighting scheme does not dynamically balance multiple attributes yet, which may lead to recommendations that prioritize one criterion over others. Future work should incorporate real user data, a larger and more diverse set of test profiles, and adaptive feature weighting to produce more balanced recommendations. With a richer dataset and more complex evaluation scenarios, differences in the performance of recommendation methods are also expected to become more apparent.

B. Comparative Analysis

To evaluate the proposed platform conceptually, a comparative analysis was conducted between the proposed system and conventional scholarship and competition information platforms. The comparison focuses on information integration, personalization capability, recommendation relevance, and strategic guidance features that support students in planning their academic and non-academic development.

TABLE I
COMPARISON BETWEEN CONVENTIONAL PLATFORMS AND THE PROPOSED SYSTEM

Feature	Conventional Platforms (Existing)	Proposed Platform (This research)
Scholarship Aggregation	Partial	Yes
Competition Aggregation	Limited	Yes
Ai-Based Recommendation	No	Yes
Personalized Fituring	Limited	Yes
Opportunity Strategy Guidance	No	Yes
Real-Time Information Update	Limited	Yes
User Profile Matching	Basic	Yes
Integration with Organizer API	Limited	Yes
Target Users (SMP & SMA)	Not Spesific	Spesific Focus

As presented in Table I, most conventional platforms primarily focus on information dissemination and typically provide only limited filtering features. In contrast, the proposed platform integrates scholarship and competition information within a centralized ecosystem while incorporating AI-based personalization mechanisms. The implementation of BM25 retrieval also improves the relevance-ranking process by considering document length normalization and term importance simultaneously.

The implementation of recommendation features based on student profiles enables the system to provide more relevant opportunities according to users' interests, talents, and educational levels. Furthermore, the inclusion of the opportunity strategy suggestion feature distinguishes the proposed system from existing platforms by supporting long-term portfolio planning rather than solely displaying available opportunities.

This comparative analysis demonstrates that the proposed platform not only improves information accessibility but also introduces a more adaptive and student-oriented recommendation approach.

Evaluation Status and Data Limitations

While the project is in the synchronization phase, an evaluation framework has been established. Formal data collection will involve testing the system with target user groups to measure recommendation accuracy and system stability. Future drafts will provide empirical results and statistical feedback once the user testing phase is finalized.

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